



BimmerStein Tuning Suite

ECU Calibration and Data Logging

USER MANUAL

Version 0.1.0 Beta 1 - Windows x64

Read the safety section before editing a calibration. Keep an untouched backup of every original file.



Safety first

Read this section before editing a calibration.

- Keep an untouched backup of every original BIN file.
- Work on a copy and use **Save As** for the first edited version.
- Confirm that the selected definition matches the exact ECU and software version.
- Review changed values, axes, units, scaling, addresses, and checksum status before saving.
- Verify the completed file with an independent method before using external flashing software.
- Make one controlled change at a time and keep notes that identify each saved revision.

Important: BimmerStein Tuning Suite does not flash or write to an ECU. It edits calibration files on disk and reads live ECU data. Flashing is a separate operation performed with other software and carries its own risk.

Definition and logger XML files are user-supplied and are not bundled with the application. A wrong or incomplete definition can display or modify the wrong bytes even when the application is working as designed.

Beta scope

Beta 1 is intended for testing and feedback. DS2 polling has been exercised on hardware, but more ECU versions, interfaces, Windows systems, and display-scaling combinations still need validation. Check multi-byte logger channels carefully because a channel definition may need explicit byte order information.

Not implemented in Beta 1: ECU flashing, Subaru SSM, generic OBD-II or ELM327, J2534, and Bluetooth transports.

Install and start

Windows installer

1. Download the Beta 1 setup executable from the project release page.
2. Run the installer and choose the destination folder.
3. Start **BimmerStein Tuning Suite** from the Start menu or desktop shortcut.

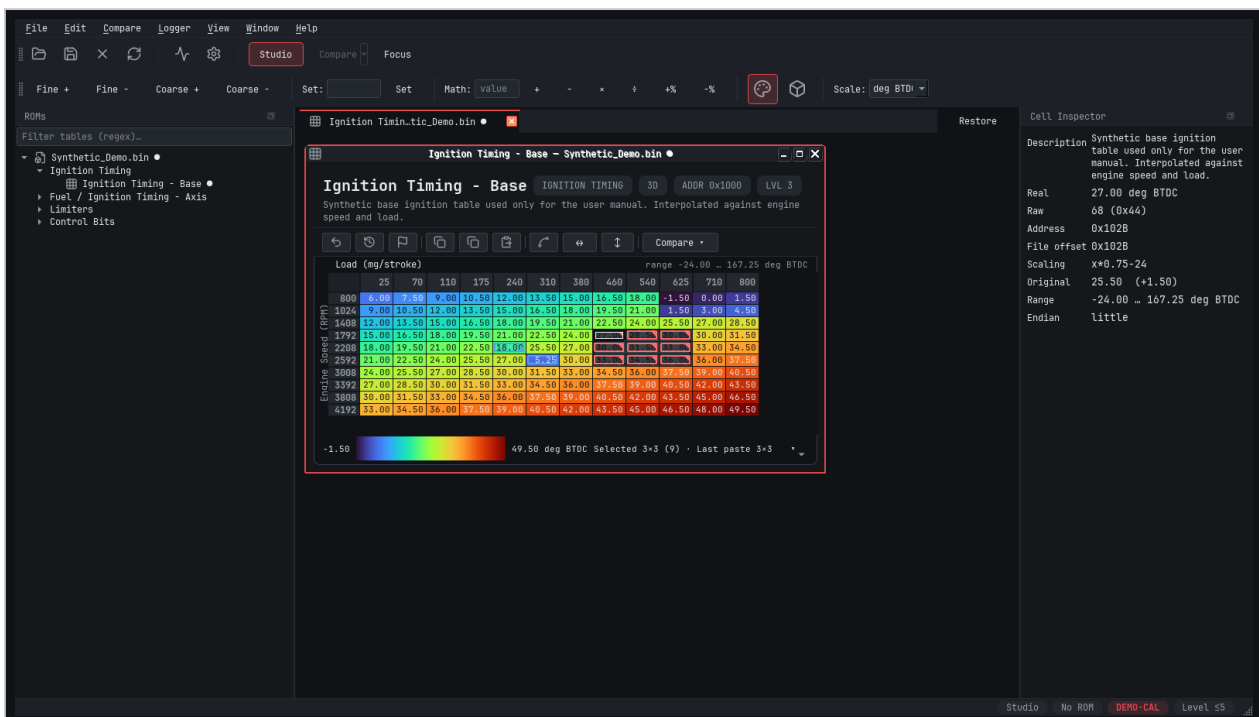
The Beta 1 executable is not code-signed. Windows may show an unknown-publisher warning. Confirm that the filename and SHA-256 checksum match the release before continuing.

Portable package

1. Download the portable Windows ZIP.
2. Extract the complete archive to a writable folder.
3. Run BimmerStein-Tuning-Suite.exe from the extracted folder.

Do not move the executable away from the DLLs and resources beside it. The portable executable is not a standalone single-file program.

Main workspace



Editor overview with a synthetic calibration

The main workspace contains the ROM tree at left, the internal table-window canvas in the center, and the Cell Inspector at right. The top toolbars provide workspace modes, edit operations, color control, 3D display, and unit selection. The status bar summarizes the active ROM and definition.

Configure definition files

Definition files tell the editor how names, addresses, storage types, axes, scaling, ranges, and descriptions map to calibration bytes.

1. Open **File > Definition Manager**.
2. Select **Add** and choose one or more compatible definition XML files.
3. Drag files into priority order. The highest compatible definition is evaluated first.
4. Confirm that each entry parses successfully, then select **Apply** or **OK**.
5. Close and reopen a ROM if it was already open. Definition changes apply to newly opened ROMs.

The application remembers configured paths. If a file is moved or renamed, update its entry in the Definition Manager.

When no definition matches

The normal open path matches raw identification bytes from the ROM against the configured definitions. If nothing matches, the editor can offer a force-load list of available XML IDs. Force loading is a diagnostic escape hatch, not proof of compatibility. Use it only when you have independently confirmed the correct definition.

What to verify after loading

- ROM or calibration ID and file size.
- Definition name or XML ID.
- Table names and categories expected for that software version.
- Plausible axes, values, units, and ranges.
- Addresses and scaling shown by the Cell Inspector.

Open and navigate a ROM

1. Select **File > Open ROM** or press **Ctrl+O**.
2. Choose a `.bin` or supported `.hex` file.
3. Confirm the ROM identification and definition status in the tree and status bar.
4. Expand the required category, then double-click a table or parameter.

The parameter tree starts collapsed so large definitions remain manageable. Use the filter field to show only names matching a regular expression. Press **Ctrl+K** to open **Go to Table**, which searches visible tables across open ROMs.

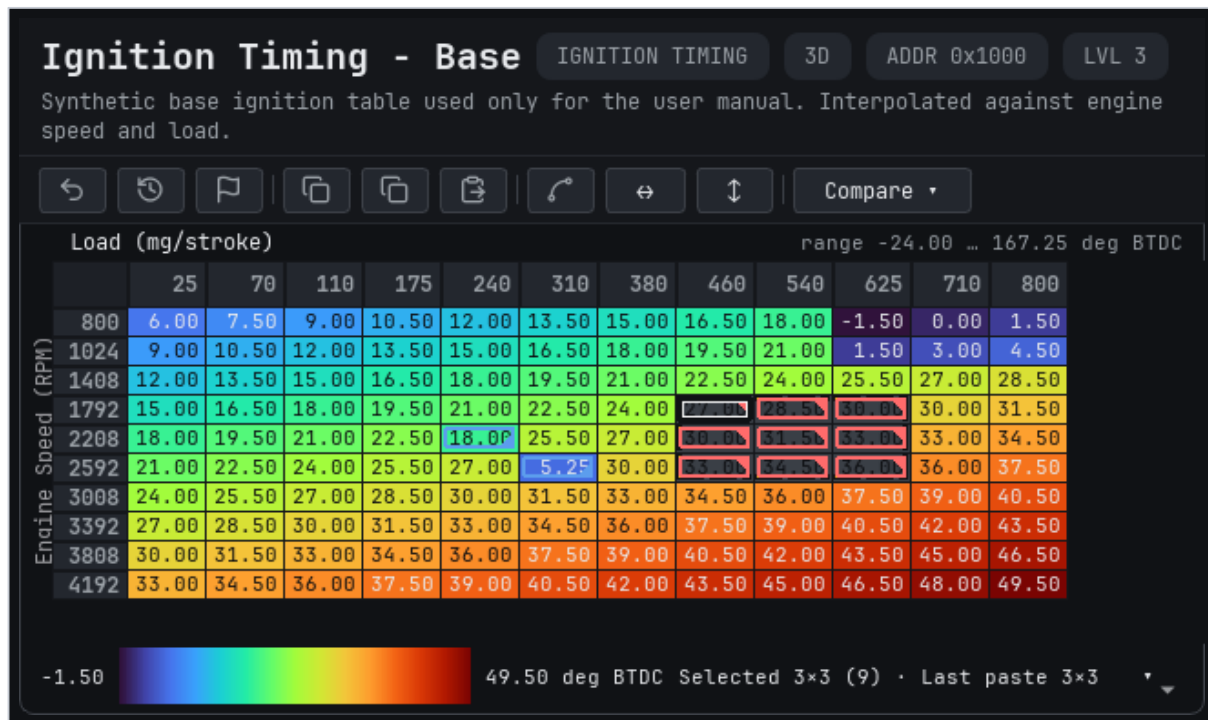
Internal windows

Each table, scalar, switch, or 3D view opens as a movable internal window. Opening another table does not reset the positions of existing windows.

- **Studio** keeps manually arranged windows.
- **Compare** places two chosen windows side by side.
- **Focus** maximizes the active internal window. Press **Esc** to restore the previous workspace.
- **Window > Tile** and **Window > Cascade** arrange all open windows on demand.

Window-size and density preferences affect table content, not the outer application window. On a constrained display the editor keeps the selected size tiers distinct while fitting them inside the available workspace.

Understand a table window



Calibration table anatomy

A grid table contains:

1. **Header** - name, category, dimensional type, storage address, and definition description.
2. **Verb bar** - undo, revert, copy, paste, interpolation, and comparison actions.
3. **Axis headers** - calibrated breakpoints for columns and rows.
4. **Value grid** - calibrated cell values colored by the selected heatmap.
5. **Range legend** - the displayed minimum, maximum, and units.

The Cell Inspector follows the current cell and can show its real value, raw bytes, storage address, file offset, scaling expression, original value, range, and byte order. Table and parameter headers also surface definition descriptions and addresses when available.

Color is a navigation aid, not a validity check. Always evaluate numbers and units directly.

Edit calibration tables

Direct entry

1. Double-click a cell, or select it and begin typing.
2. Enter the required calibrated value.
3. Press **Enter** to commit or **Esc** to cancel.

The editor converts the real value to the table's storage representation, then displays the value that can actually be encoded. Values outside the defined range are rejected or constrained by the table definition.

Select and adjust regions

- Click one cell to make it current.
- Drag across cells or use standard Ctrl/Shift selection to select a region.
- Use **Fine +**, **Fine -**, **Coarse +**, or **Coarse -** for definition-sized steps.
- Enter a number beside **Set** and apply it to the selection.
- Choose a math operation to add, subtract, multiply, divide, or apply a percentage.
- Select **Interp**, horizontal interpolation, or vertical interpolation where appropriate.

Visual edit indicators

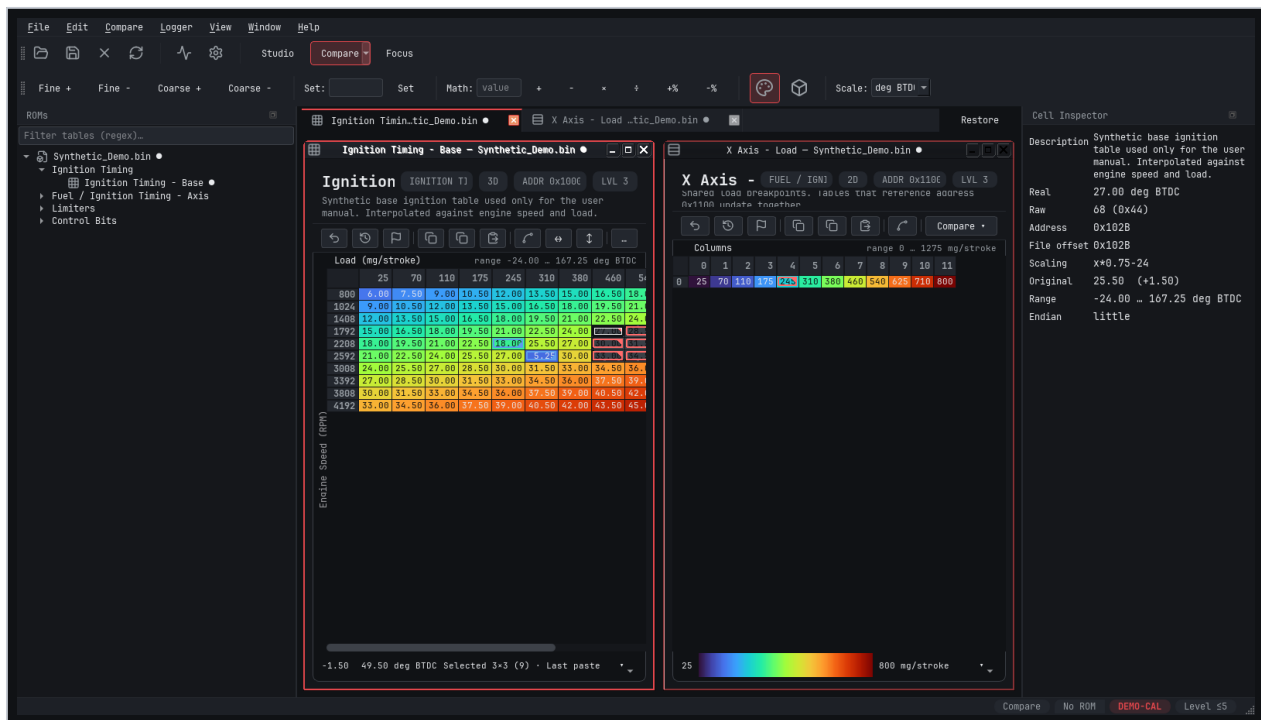
- A completed selection uses a neutral gray overlay.
- The region being dragged uses a blue overlay.
- The current cell has a high-contrast double outline.
- A changed value has a red border and corner marker when it increased.
- A changed value has a blue border and corner marker when it decreased.
- A recent paste has an amber dashed perimeter and a size summary.

These indicators compare the current data with the active revert baseline. They do not alter the heatmap values.

Copy and paste

Copy Sel copies the selected rectangle as tab-separated values. **Copy Table** copies the full grid. Paste begins at the current cell and supports rectangular data copied from compatible editors or spreadsheets. Confirm dimensions, axis orientation, units, and rounding after every paste.

Edit shared table axes



Shared physical axis editing

Storage-backed axis headers are editable. Double-click a column or row header, enter the calibrated breakpoint, and press **Enter**.

Some definitions expose the same breakpoint data as a separate axis table. If multiple open tables refer to the same shared physical axis bytes, an edit is synchronized across all of those views. This prevents one window from displaying stale breakpoints after another window changes them.

Before editing an axis:

- Confirm the axis name, units, address, size, storage type, and scaling.
- Identify every table that shares the physical data.
- Preserve the ordering required by the calibration strategy.
- Recheck every affected table after the change.

A static or label-only axis is descriptive and cannot be edited. The editor only commits axis changes when the definition maps that axis to writable storage.

Work with scalars and switches



Scalar and switch parameter windows

Scalar parameters

Scalar windows present one large calibrated value, its units, and four step buttons. Double-click the value to enter a number directly, or use the fine and coarse decrement/increment buttons. The definition controls storage, scaling, formatting, and step size.

Switch parameters

Switch windows list the named states and the raw byte pattern for each state. Selecting a state writes its defined bytes to the open ROM image. **Undo All** returns the parameter to its revert baseline.

Bitwise switches expose independent named flags when the definition maps several options into one stored value.

Addresses and descriptions

The title/header and Cell Inspector expose available definition metadata for tables, scalars, switches, and axes. For switches, review the storage address plus the byte pattern shown beside each state. Descriptions come from the loaded definition and may be absent when the source XML does not provide one.

Compare, undo, and save

Undo

- **Undo** or **Ctrl+Z** reverses the most recent edit in the active table.
- **Undo All** or **Ctrl+Shift+Z** restores the active table to its current revert baseline.

Set Revert Point

Set Revert Point accepts the current table values as a new local baseline. After setting it, change borders clear and **Undo All** returns to that new point instead of the values first loaded from disk. This is useful for comparing a new experiment with an already accepted set of edits.

Setting a revert point does not save the ROM file. It also does not erase the original file on disk.

Compare

Use the table **Compare** menu to show changes or compare the active table against a compatible table. **Compare > Compare Images** compares open ROM images. Use **Window > Compare Active With** to choose two open views for the side-by-side workspace.

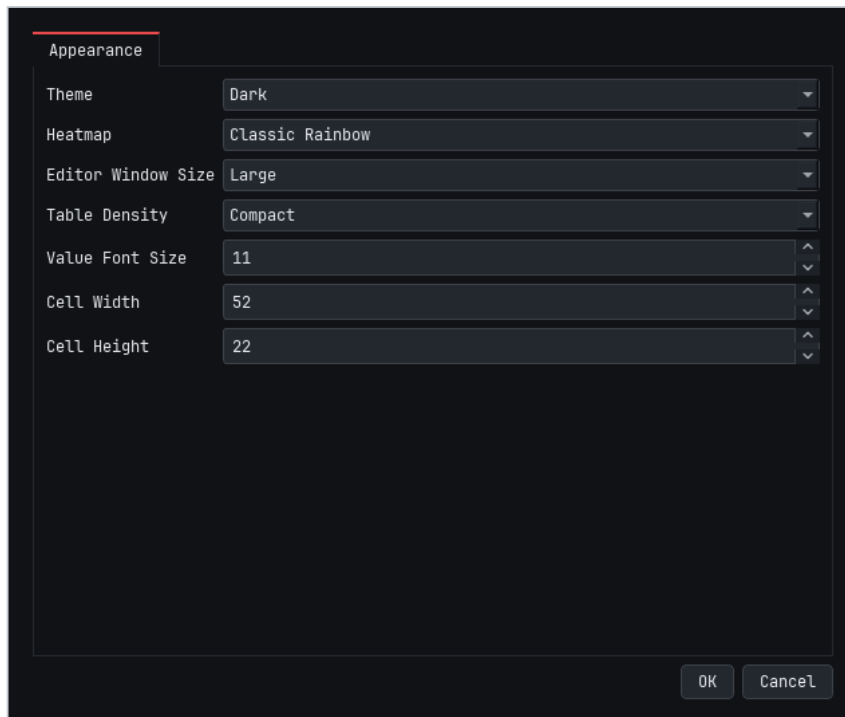
Save safely

1. Review all changed tables and shared axes.
2. Select **File > Save As** or press **Ctrl+Shift+S** for a new revision.
3. Choose a descriptive filename that preserves the original.
4. Review checksum and status information.
5. Reopen and independently verify the saved file before any separate flashing operation.

Save writes to the current file path. The editor warns about unsaved ROM changes when closing.

Appearance and table sizing

Open **Edit > Settings** to change:



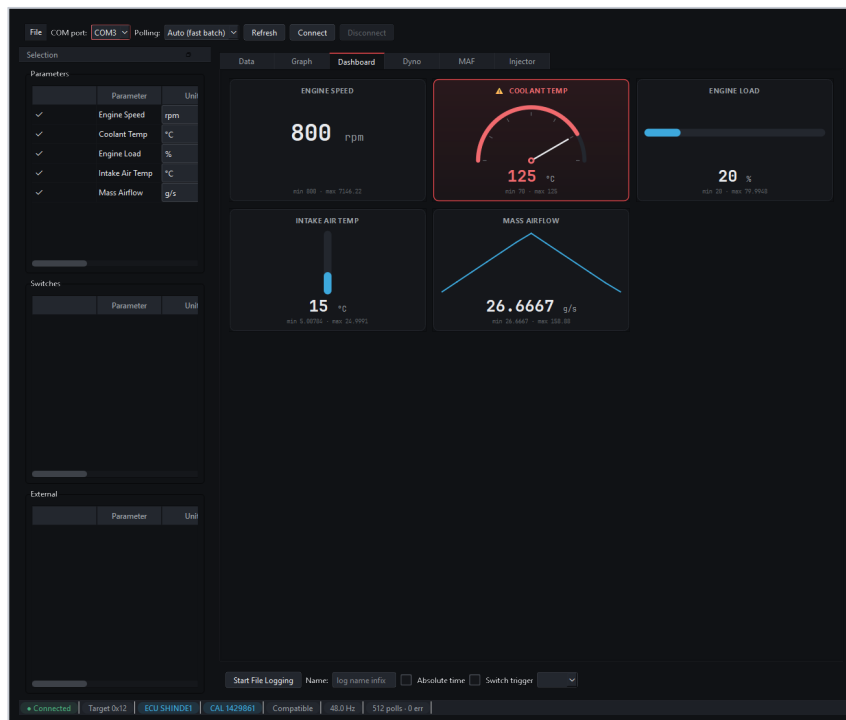
Appearance and table sizing settings

- Theme: Dark, Light, or System.
- Heatmap: Rainbow or Viridis.
- Editor window size: Small, Medium, or Large.
- Table density: Normal or Compact.
- Numeric value font size.
- Preferred cell width and height.

The editor calculates natural column widths from the actual formatted values and axis labels, then fits the internal window to its content within the selected size limit. Small tables should remain compact instead of inheriting unused space from large maps. Very large tables use scrollbars when the complete natural size cannot fit on the current display.

If text is cropped, first try Normal density or a smaller value font. Also confirm the Windows display-scaling percentage and restart the application after a major monitor or scaling change.

Configure and use the logger



Logger dashboard using synthetic replay data

The logger reads live values through a supported communication transport. It does not change ECU calibration memory.

Before connecting

1. Connect a supported serial interface and install its driver.
2. Open the Logger with **Logger > Launch Logger** or **Ctrl+L**.
3. Load or configure the required user-supplied logger definition.
4. Select the COM port and refresh the list if necessary.
5. Choose a polling mode.

Auto (fast batch) uses the fastest eligible DS2 batch/grouped reads. **Compatible** uses individual memory reads and is the safer fallback when a definition or ECU does not behave correctly in batch mode.

Select channels

The Selection dock separates Parameters, Switches, and External channels. Each row can be enabled for polling and independently routed to Data, Graph, or Dashboard.

Start with a small channel set. Adding channels increases the amount of work in each polling cycle and can reduce update rate. Confirm units, plausible values, and stable error counts before adding more.

Select **Connect**. The status bar reports connection state, target module, ECU ID, calibration ID, polling mode, rate, polls, errors, and recording state.

Logger views and recording

Data

The Data tab provides a compact numerical view of selected live channels. Use it for validation and when maximum information density matters.

Graph

The Graph tab plots recent channel history. Keep scale differences in mind when interpreting several signals together.

Dashboard

The Dashboard tab presents configured digital, needle, bar, and sparkline gauges. Warning thresholds and gauge styles are presentation aids and must be configured for the channel units.

Dyno and analysis

The Dyno tab can capture and analyze a suitable acceleration run using configured vehicle data. Analysis tabs may provide additional workflows when their required channels are selected. Results depend on the accuracy of the logger data, vehicle profile, environmental inputs, and test method; they are estimates rather than chassis-dyno measurements.

CSV recording

1. Use **File > Set Log Directory** to choose the output folder.
2. Optionally enter a short log-name suffix.
3. Select **Start File Logging**, or press **F1**.
4. Perform the test while monitoring connection and error status.
5. Stop logging before disconnecting.

Switch-triggered logging can arm recording around a defined switch transition. Confirm the selected switch and trigger behavior in a safe stationary test before relying on it.

Logger profiles save channel/view selections, port, and log directory for reuse. They do not embed the ECU definition itself.

Keyboard shortcuts

Area	Shortcut	Action
Application	Ctrl+O	Open ROM
Application	Ctrl+S	Save
Application	Ctrl+Shift+S	Save As
Application	Ctrl+W	Close ROM
Application	F5	Refresh active ROM
Application	Ctrl+L	Launch Logger
Application	Ctrl+K	Go to Table
Workspace	Ctrl+Alt+1	Studio mode
Workspace	Ctrl+Alt+2	Compare mode
Workspace	Ctrl+Alt+3	Focus mode
Workspace	Ctrl+F4	Close active internal window
Workspace	Ctrl+Tab	Next internal window
Workspace	Ctrl+Shift+Tab	Previous internal window
Table	Ctrl+Z	Undo last table edit
Table	Ctrl+Shift+Z	Undo all to revert point
Table	Ctrl+C	Copy selection
Table	Ctrl+Shift+C	Copy table
Table	Ctrl+V	Paste
Table	Shift+I	Interpolate selection
Table	Shift+H	Horizontal interpolation
Table	Shift+V	Vertical interpolation
Table	+ / _	Coarse increment / decrement
Table	*	Multiply selection
Logger	F1	Start or stop file logging
Logger	F9	Unselect all channels
Logger	F11	Toggle Selection dock
Logger	F12	Cycle gauge style

Shortcuts for table editing apply when the active grid or one of its child controls has focus.

Troubleshooting

The ROM has no matching definition

- Confirm that definition paths are present and parse successfully.
- Confirm the ROM file size and identification bytes.
- Use a definition for the exact software version.
- Do not force load an approximate definition merely because its table list looks familiar.

Values or axes look implausible

- Stop editing and compare with the untouched backup.
- Verify storage address, storage type, byte order, scaling, and units in the inspector.
- Confirm that the correct inherited definition resolved.
- Reopen the file after correcting the definition.

A table does not fit or text is cropped

- Try Compact density, a smaller value font, or the Small window tier.
- For clipped labels, try Normal density so row height follows the font.
- Check Windows display scaling and available workspace size.
- Use Focus mode for very large tables.

Logger will not connect

- Confirm the correct COM port and interface driver.
- Close other software that may own the port.
- Verify ECU power and the communication wiring.
- Try Compatible polling mode.
- Confirm that the logger definition targets the correct ECU module and address class.

Live values are wrong but polling is stable

- Verify channel address, byte length, storage type, byte order, expression, and units.
- Do not assume the global byte-order default is correct for a multi-byte channel.
- Compare against a trusted capture or known operating condition.

The application closes with unsaved changes

Use the close prompt carefully. Choose to keep working if you have not saved or intentionally discarded all changes. Revert points are in-memory edit baselines and are not substitutes for saved files.

Pre-save checklist

- [] The original BIN remains untouched and separately backed up.
- [] The definition exactly matches the ECU/software version.
- [] Every changed value and shared axis was reviewed in calibrated units.
- [] Unexpected changed-cell indicators were investigated.
- [] The output uses a new, descriptive filename.
- [] Checksum and status information were reviewed.
- [] The saved file was reopened and independently verified.
- [] Any later flashing process has its own recovery plan and verified power supply.

Support and project information

- Project: github.com/CAATZ/bimmerstein-tuning-suite
- Issues: [Report a bug or request a feature](#)
- Release notes: Beta 1 release notes
- Licensing: GNU GPL and third-party notices

Useful bug reports include the ECU or ROM version, Windows version, display-scaling percentage, application theme, definition version, affected table or parameter, exact reproduction steps, and screenshots. Do not attach proprietary or personal files unless you intend to share them.

BimmerStein Tuning Suite is independent software. It supports compatible public formats and workflows but is not a Git fork of RomRaider. See the third-party notices for retained credits, source references, and licenses.

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